

THE DETERMINISTIC FOUR-TIER FALLBACK ESCALATION LADDER

Graded Safety Authority: Escalating Physical Cost, Decreasing Software Dependency & Increasing Isolation

1 QP Safety Filter Least-Squares Projection

TRIGGER CONDITION

Policy proposal u_{nom} violates CBF;
Feasible set non-empty ($U_{safe} \neq \emptyset$).

MECHANISM & ALGORITHM

Orthogonal QP projection in SRAM;
Calculates minimal correction $\Delta u = -\lambda a$.

REACTION LATENCY

$\leq 150 \mu s$ (Active-set QP @ 1 kHz)

PHYSICAL & SYSTEM COST

- 0 mechanical wear / 0 brake wear
- 0 mission downtime (Active Task)

REVERSIBILITY

100% Reversible (Continuous)

EXECUTION SUBSTRATE

NPU/MPU Software / MCU
Pre-allocated static SRAM arrays

STANDARD / CLASSIFICATION

Control Barrier Certificate (CBF)

2 Active Position Hold IEC 60204-1 Category 2 (SS2)

TRIGGER CONDITION

Empty safe set ($U_{safe} = \emptyset$), solver fail,
or upstream policy lease timeout.

MECHANISM & ALGORITHM

PID/LQR arrests motion to x_{hold} ;
Continuous PWM active current control.

REACTION LATENCY

10 – 50 ms (Closed-loop holding)

PHYSICAL & SYSTEM COST

- Continuous Joule heating (I^2R)
- 0 brake wear; mission paused

REVERSIBILITY

Reversible on lease renewal

EXECUTION SUBSTRATE

Real-Time Core / RTOS Loop
Deterministic current regulation

STANDARD / CLASSIFICATION

IEC 60204-1 Category 2 (Safe Stop 2)

3 Dynamic Decel Stop IEC 60204-1 Category 1 (SS1)

TRIGGER CONDITION

Tracking error breach ($|e| > \epsilon_{max}$),
obstacle approach, lease expiry.

MECHANISM & ALGORITHM

Certified max dynamic braking a_{max} ;
Brake chopper absorbs DC-bus surge.

REACTION LATENCY

100 – 500 ms (Stopping profile)

PHYSICAL & SYSTEM COST

- Peak torque load on gear teeth
- Mission aborted; requires re-homing

REVERSIBILITY

Non-Reversible (Task Abort)

EXECUTION SUBSTRATE

Dedicated Safety MCU (Cortex-R)
Independent sensor channel & power

STANDARD / CLASSIFICATION

IEC 60204-1 Category 1 (Safe Stop 1)

4 Safe Torque Off (STO) IEC 60204-1 Category 0 (STO)

TRIGGER CONDITION

Watchdog timeout ($>50 ms$), shoot-through,
overcurrent comparator, E-stop switch.

MECHANISM & ALGORITHM

Optocouplers drop gate bias ($<5 \mu s$);
Spring-applied friction brakes clamp.

REACTION LATENCY

$< 5 \mu s$ (Silicon) / 15 ms (Pad seat)

PHYSICAL & SYSTEM COST

- Friction pad ablation & shock stress
- Potential gravity sag before pad seat

REVERSIBILITY

Manual Lockout Reset

EXECUTION SUBSTRATE

Galvanic Hardware Safety Relay
Hardware optocouplers & contactors

STANDARD / CLASSIFICATION

ISO 13849-1 PLe / ISO 26262 ASIL-D

← Low Mechanical Cost / High Reversibility

ESCALATION OF DEFENSIVE INTERVENTION

High Hardware Cost / Non-Reversible →

SUBSTRATE INDEPENDENCE:

Shared MPU / RTOS Core

→ Isolated Safety Co-Processor

→ Dual-Channel Safety MCU

→ Galvanic Analog Hardware / E-Stop

Systems Architectural Rule: Lower rungs must execute on substrates physically isolated from upstream software failure modes.